MATH2012 Fundamental Concepts of Mathematics Notes

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Introduction

This is an introductory mathematics course. In high school, mathematics appears to be calculations. In undergraduate-level mathematics, however, things are more "proof"-based. This course aims to help students transition from high school mathematics to university mathematics.

This note should be a revision note before tests or exams but not a starting material. You are strongly advised to read the following textbook before reading this note.

Recommended reading: Gary Chartrand, Albert D. Polimeni, Ping Zhang: *Mathematical Proofs: A Transition to Advanced Mathematics* (Pearson, 2012, Third Edition)

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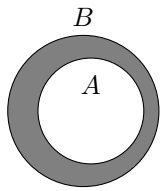
1 Basic Set Theory

We say that A is a subset of B (in symbol, $A \subset B$) if all elements in A are in B. In Venn Diagram:



Example 1.1. The set of square numbers (0, 1, 4, 9, etc) is a subset of the set of non-negative integers (0, 1, 2, 3, 4, 5, 6, etc). But the set of square numbers is not a subset of even numbers (0,2,4,6,8, etc (usually -2,-4,-6, etc are also included)) because 1 is a square number that is not even.

By definition, A is a subset of A itself. We say that A is a proper subset of B if A is a subset of B but B is not a subset of A.

To prove two sets X and Y are equal rigorously, we have to prove $X \subset Y$ and $Y \subset X$.

Example 1.2. (Please understand the definitions of the symbols first before you read this example, the same for all following examples) Prove that the following sets are equal:
$A = \{2a + 1 : a \in \mathbb{Z}\}$
$B = \{x^2 - y^2 : x, y \in \mathbb{N}, \frac{x+y}{2} \notin \mathbb{Z}\}$
Solution. Obviously, we have $B \subset A$ because every element in B can be written as $x^2 - y^2$ with $x, y \in \mathbb{N}$ and exactly one of x and y being odd and the other being even, and in any case $x^2 - y^2$ must be odd and hence in A.

To show that $A \subset B$, note that every element in A can be written as $2a + 1$ for some $a \in \mathbb{Z}$ which can be written as $(a+1)^2 - a^2$, and is therefore an element in B.

Thus $A = B$.

2 Logic

You will frequently see words like "there exist(s)", "for every/all/any", "if", "only if", "if and only if", "the converse" etc in future mathematics courses. Let's use some not-so-mathematics examples to illustrate what these words mean.

If you heard from the course instructor that "**There exists** a student that failed MATH2012", it means that at least one student failed MATH2012, and it does not say exactly one student failed the course in the broad sense.

If you heard from the course instructor that "**For all** students who score 80 or above, you will receive an A+", then you will get an A+ if you know your score is 85. But if you score lower than 80, say you score 75, then what the instructor said concerns nothing about you. This is because he/she just

guaranteed you an A+ if you score 80 or more and doesn't mean you can't get an A+ if you score 75.

If the course instructor said "**If** you score 80 or above, you will receive an A+", it has the same meaning as the above sentence.

If the course instructor said "You will get an A+ **if and only** if you get 80 or above", then it means that you will not get an A+ if you score lower than 80 (and get one if you score 80 or above).

Suppose there is a statement "If A, then B". Then the **converse** of this statement is "If B, then A". For instance, if the teacher said "If you score 80 or above, you will get an A+", then the **converse** of what the teacher said is "If you get an A+, you score 80 or above".

3 Mathematical Induction

You might have learned this in high school. Here we give an example that might not appear in high school mathematics textbooks.

Example 3.1. Prove by induction that every integer greater than 11 can be written as $4x + 5y$ where x and y are non-negative integers.
Solution. In high school, we mostly apply mathematical induction in this way: "The statement is true for $n = 1$; if the statement is true for $n = k$, then we prove that the statement is also true for $n = k + 1$ and so the statement is true for all $n \in \mathbb{N}$". However, the induction hypothesis does not work in this example. Indeed, you should observe $12 = 4 \times 3 + 5 \times 0$, $13 = 4 \times 2 + 5 \times 1$, $14 = 4 \times 1 + 5 \times 2$, $15 = 4 \times 0 + 5 \times 3$ so the statement $S(n)$: "n can be written as $4x + 5y$, x and y are non-negative integers" is true for $n = 12, 13, 14, 15$.

If $S(n)$ is true for some integer $n > 11$, then $n = 4x + 5y$ for some non-negative integers x and y. So, $n + 4 = 4(x + 1) + 5y$ and $S(n + 4)$ is also true. This completes the induction steps and $S(n)$ is true for all integers $n > 11$.

4 Equivalence Relation

A binary relation $\sim$ on a set X is said to be
reflexive if $x \sim x$ for all $x \in X$;
symmetric if $x \sim y$ implies $y \sim x$ for all $x, y \in X$;
transitive if $x \sim y$ and $y \sim z$ imply $x \sim z$ for all $x, y, z \in X$.

A reflexive, symmetric and transitive relation $\sim$ on X is called an equivalence relation.

Example 4.1. Consider the relation $>$ on $\mathbb{R}$, it is a transitive relation because $x > y$ and $y > z$ imply $x > z$ for $x, y, z \in \mathbb{R}$. However, it is not a reflexive nor symmetric relation (and thus not an equivalence relation) on $\mathbb{R}$.

Example 4.2. Consider the relation $=$ on $\mathbb{R}$, it is a reflexive relation because $x = x$ for all $x \in \mathbb{R}$; it is a symmetric relation because $x = y$ implies $y = x$ for

all $x, y \in \mathbb{R}$; it is also a transitive relation because $x = y$ and $y = z$ imply $x = z$ for all $x, y, z \in \mathbb{R}$. Thus $=$ is an equivalence relation on $\mathbb{R}$.

5 Functions

By a function $f : A \to B$, we mean a map sending each element in A to an element in B. A function $f : A \to B$ is injective if $f(a) = f(b)$ if and only if $a = b$. A function $f : B \to A$ is surjective if (in symbol $B = f(A)$). A function $f : A \to B$ is bijective if it is both injective and surjective.

Example 5.1. $f : \mathbb{N} \to \mathbb{N}$ ($2\mathbb{N}$ means the set of positive even numbers) defined by $f(x) = 2x$ is injective. Since $1 \neq 2x$ for any $x \in \mathbb{N}$, 1 is not an image of f and hence f is not surjective. To modify f so that it becomes a bijective function, we can take $f : \mathbb{N} \to 2\mathbb{N}$ ($2\mathbb{N}$ means the set of positive even numbers), $f(x) = 2x$.

Example 5.2. Let S be a nonempty set and $\mathcal{P}(S)$ the power set of S. For any $A, B \in \mathcal{P}(S)\backslash\{\emptyset\}$, define a relation R on $\mathcal{P}(S)\backslash\{\emptyset\}$ by ARB if there exists a bijective function $f : A \longrightarrow B$. Show that R is an equivalence relation on $\mathcal{P}(S)\backslash\{\emptyset\}$.
Solution. Firstly, R is reflexive: for any $A \in \mathcal{P}(S)\backslash\{\emptyset\}$, $id_A : A \to A$, $x \mapsto x$ is a bijective function, so ARA.

Secondly, R is symmetric: for all $A, B \in \mathcal{P}(S)\backslash\{\emptyset\}$, if ARB, then there is a bijective $f : A \to B$, and $f^{-1} : B \to A$ is a bijective map from B to A, so BRA.

Finally, R is transitive: for all $A, B, C \in \mathcal{P}(S)\backslash\{\emptyset\}$, if ARB and BRC, then there is a bijective $f : A \to B$ and a bijective $g : B \to C$, so $g \circ f : A \to C$ is a bijective map from A to C, so ARC.

6 Cardinalities of Sets

This topic is perhaps the signature of the course. When we talk about "infinity", we know that there are infinitely many integers, infinitely many even numbers, infinitely many square numbers, infinitely many prime numbers, infinitely many real numbers .....

It is natural to think that some "infinity" is bigger than some "infinity", but it might be hard to make a reasonable comparison without reading the textbook. After reading the textbook, we get an idea of how to define when two sets have equal sizes using the notion of bijective functions - two sets A and B have equal sizes if there is a bijective mapping $f : A \to B$.

We now state some important theorems and facts without (or without detailed) proof.

Example 6.1. $k\mathbb{N} \subset \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$ for all $k \in \mathbb{N}$. All four sets are of the same sizes - they have countably infinite elements (countable means the set is either finite or has equal size as $\mathbb{N}$).

Example 6.2. $\mathbb{R}$ and $\mathbb{R}\backslash\mathbb{Q}$ have an uncountable number of elements and their sizes are greater than that of $\mathbb{N}$ (i.e. there are injective maps from $\mathbb{Q}$ to $\mathbb{R}$ and from $\mathbb{Q}$ to $\mathbb{R}\backslash\mathbb{Q}$, but no bijective maps from $\mathbb{Q}$ to $\mathbb{R}$ nor from $\mathbb{Q}$ to $\mathbb{R}\backslash\mathbb{Q}$).

Example 6.3. For any set S, no matter S is countable or uncountable, we have $|S| < |2^S|(= |\mathcal{P}(S)|)$.

Theorem 6.1 (Schröder-Bernstein Theorem)**.** If A and B are sets and there are injective maps $f : A \to B$ and $g : B \to A$, then $|A| = |B|$.

Example 6.4. $|2^{\mathbb{N}}| = |\mathbb{R}|$.
Solution. Note that $|\mathbb{R}| = |(0,1)|$ by consider the bijective map $f : \mathbb{R} \to (0,1)$, $f(x) = \frac{\tan^- 1x}{\pi} + \frac{1}{2}$. So, it suffices to show $|2^{\mathbb{N}}| = |(0,1)|$.

First, note that the map $g : 2^{\mathbb{N}} \to (0,1)$ by $g(a_1, a_2, a_3, \cdots, a_n, \cdots) = 0.a_1a_2a_3\cdots a_n\cdots$, $a_n \in \{0,1\}$ for all $n \in \mathbb{N}$ is an injective map because $g(a_1, a_2, a_3, \cdots, a_n, \cdots) = g(b_1, b_2, b_3, \cdots, b_n, \cdots)$ if and only if $a_n = b_n$ for all $n \in \mathbb{N}$.

Second, to construct an injective map from $(0,1)$, it should be aware things like $0.4\dot{9} = 0.5$. Luckily, it can be observed easily that each number in $(0,1)$ can be expressed uniquely by an infinite sum $\sum_{i=1}^{\infty} \frac{a_i}{2^i}$ with each $a_i \in \{0,1\}$ with no $k \in \mathbb{N}$ such that $a_n = 1$ for all $n \geq k$. After using this expression, we consider the map $h : (0,1) \to 2^{\mathbb{N}}$, $h(\sum_{i=1}^{\infty} \frac{a_i}{2^i}) = (a_1, a_2, \cdots, a_n, \cdots)$, then h is an injective map.

By Schröder-Bernstein Theorem, $|2^{\mathbb{N}}| = |\mathbb{R}|$.

Example 6.5. $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$.
Solution. Firstly, the map $f : \mathbb{N} \to \mathbb{N} \times \mathbb{N}$, $n \mapsto (n, 1)$ is clearly an injective map.

Secondly, the map $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, $(a, b) \mapsto \frac{(a+b-1)(a+b-2)}{2} + b$ is injective. To see this, assume $\frac{(a+b-1)(a+b-2)}{2} + b = \frac{(c+d-1)(c+d-2)}{2} + d$. If $a + b > c + d$, we have

$$\begin{aligned}
\frac{(a+b-1)(a+b-2)}{2} + b &\geq \frac{(c+d)(c+d-1)}{2} + b \\
&= \frac{(c+d-1)(c+d-2)}{2} + c + d - 1 + b \\
&> \frac{(c+d-1)(c+d-2)}{2} + d
\end{aligned}$$

which means $g(a, b) > g(c, d)$, contradiction. Similarly, we cannot have $c + d > a + b$. This forces $a + b = c + d$. When $a + b = c + d$, $g(a, b) = g(c, d)$ implies $b = d$ and which then gives $(a, b) = (c, d)$.

By Schröder-Bernstein Theorem, $|\mathbb{N}| = |\mathbb{N} \times \mathbb{N}|$.

Remark 6.1. It should be noted that the map $(a, b) \mapsto \frac{(a+b-1)(a+b-2)}{2} + b$ is bijective, making Schröder-Bernstein Theorem not that useful in the proof. Another construction of an injective map from $\mathbb{N} \times \mathbb{N}$ makes use of the fundamental theorem of arithmetic.

Exam Questions

We use the 2020 June Exam to demonstrate how you can apply what you have learned in class to answer the exam questions.

1. Prove or disprove each of the following statements:
(a) For any nonempty sets A, B, C, D,

$$(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$$

(b) For any subset A of the set $\mathbb{I}$ of irrational numbers, if A is infinite, then A is uncountable.
(c) For any nonempty sets A, B and for any function $f : A \longrightarrow B$, if $X = f^{-1}(f(X))$ for any subset X of A, then f is injective.
(d) For any nonempty sets A, B and for any function $f : A \longrightarrow B$, if $f\left(f^{-1}(Y)\right) = Y$ for any subset Y of B, then f is surjective.
Solution. (a) False. For instance, when $A = \{1\}, B = \{2\}, C = \{3\}, D = \{4\}$, $(A \times B) \cup (C \times D) = \{(1, 2), (3, 4)\}$ while
$(A \cup C) \times (B \cup D) = \{(1, 2), (3, 2), (1, 4), (2, 4)\}$.
(b) False. For instance $A = \{n + \pi : n \in \mathbb{N}\} \subset \mathbb{I}$ is a countably infinite subset of $\mathbb{I}$.
(c) True, because if $f(x) = f(y)$, $\{x\} = f^{-1}(\{f(x)\}) = f^{-1}(\{f(y)\}) = \{y\}$.
(d) True, for any $b \in B$, $f\left(f^{-1}(Y)\right) = Y$ implies $f^{-1}(\{b\}) \neq \phi$ and $f(x) = b$ for some $x \in f^{-1}(\{b\})$. So f is surjective.
2. Consider the sequence $a_1, a_2, a_3, \ldots$ of numbers satisfying $a_1 = 4$, $a_2 = 14$ and

$$a_n = 4a_{n-1} - a_{n-2} \quad \text{for any integer } n \geq 3$$

Find a general formula for a_n and prove by induction that your answer is correct.
(Hint: Consider the quadratic equation $x^2 - 4x + 1 = 0$.)
Solution. Solve the quadratic equation $x^2 - 4x + 1 = 0$, the roots are $2 \pm \sqrt{3}$. So the formula for a_n should be $a(2 + \sqrt{3})^n + b(2 - \sqrt{3})^n$ for some constants a and b. One can easily find that $a = b = 1$, then use M.I. to prove the formula.
3. Let R be the relation on $\mathbb{R}^2$ defined by

$$(a, b)R(c, d) \text{ if } (ab = 0 \text{ and } cd = 0) \text{ or } (ac > 0 \text{ and } bd > 0)$$

(a) Prove that R is an equivalence relation.
(b) Find all the distinct equivalence classes and hence give a partition of $\mathbb{R}^2$ with respect to R.
Solution. Firstly, $(a, b)R(a, b)$ for all $(a, b) \in \mathbb{R}^2$ as if we do not have $ab = 0$, then $a, b \neq 0$ and get $a^2 > 0$ and $b^2 > 0$. So R is reflexive.
Next, R is symmetric because if we have $(a, b)R(c, d)$, then ($ab = 0$ and $cd = 0$) or ($ac > 0$ and $bd > 0$), which is the same as saying ($ba = 0$ and $dc = 0$) or ($ca > 0$ and $db > 0$), hence $(c, d)R(a, b)$.

Finally, R is transitive because if $(a,b)R(c,d)$ and $(c,d)R(e,f)$, then ($ab=0$ and $cd=0$) or ($ac>0$ and $bd>0$), and ($cd=0$ and $ef=0$) or ($ce>0$ and $df>0$). If $cd=0$, we must have $ab=0$ and $ef=0$, so $(a,b)R(e,f)$. If $cd\neq 0$, then $c,d\neq 0$, moreover, $ac>0$, $bd>0$, $ce>0$ and $df>0$, which imply $ae=\frac{(ac)(ce)}{c^2}>0$ and $bd=\frac{(bd)(df)}{d^2}>0$, so $(a,b)R(e,f)$ in this case as well.
4. Let $f:\mathbb{R}\longrightarrow\mathbb{R}$ be a function satisfying

$$f(yf(x)+y)=xy+f(y)\quad \text{for any } x,y\in\mathbb{R} \tag{*}$$

(a) Prove that f is injective. Hence show that $f(0)=0$.
(b) Prove that f is surjective.
(c) Hence, or otherwise, find all functions $f:\mathbb{R}\longrightarrow\mathbb{R}$ satisfying (*).
Solution. (a) For $x,y\in\mathbb{R}$,
$f(x)=f(y)\Rightarrow f(1\cdot f(x)+1)=f(1\cdot f(y)+1)\Rightarrow x+f(1)=y+f(1)\Rightarrow x=y$.
So, f is injective. Hence, plug $x=0$ into $(*)$, we have $f(yf(0)+y)=f(y)$ for all $y\in\mathbb{R}$, this forces $yf(0)+y=y$ and $f(0)=0$.
(b) For any $r\in\mathbb{R}$, $f(f(r-f(1))+1)=(r-f(1))\cdot 1+f(1)=r$, so f is surjective.
(c) Assume $y\neq 0$, plug $x=-\frac{-f(y)}{y}$ into $(*)$, we get $f(yf(\frac{-f(y)}{y})+y)=0$. By the injectivity of f, we must have $yf(\frac{-f(y)}{y})+y=0$ forcing $f(\frac{-f(y)}{y})=-1$ for all $y\in\mathbb{R}\backslash\{0\}$. By the injectivity of f again, we have $\frac{-f(y)}{y}=\frac{-f(1)}{1}=-f(1)$ for all $y\in\mathbb{R}\backslash\{0\}$. This means $f(y)=f(1)y$ for all $y\in\mathbb{R}$. It can be checked easily that $f(x)=mx$, $m\neq 0$ are possible.
5. (a) Prove that for any positive integer m,

$$5^m\equiv 2^m\equiv\begin{cases}1(\mod 3) & \text{if } m \text{ is even}\\ 2(\mod 3) & \text{if } m \text{ is odd}\end{cases}$$

(b) Prove that if there exist positive integers x,y,z satisfying the equation

$$3^x+4^y=5^z$$

then z must be even. Write $z=2k$ for some positive integer k. Prove that k must be odd and y must be even.
(c) Hence, or otherwise, prove that $(x,y,z)=(2,2,2)$ is the only positive integral solution of the equation $3^x+4^y=5^z$.
(Hint: Use (b) and consider modulo 8.)
Solution. (a) Can be done by M.I. easily.
(b) Consider modulo 3, we get immediately from the equation and (a) that z must be even. Consider modulo 4, we get immediately that x is even. Then write $z=2k$ and $x=2t$, we get $(5^k-3^t)(5^k+3^t)=4^x=2^{2x}$, by fundamental theorem of arithmetic, we must have $5^k-3^t=2^a$ and $5^k+3^t=2^b$ for some $a,b\in\mathbb{N}$ such that $a+b=2x$, which means $2\cdot 5^k=2^a+2^b$, which means one of a and b equals 1 by comparing the exponents of 2 in the prime factorization of both sides. Then, we get $5^k-3^t=2$, and consider modulo 8, we get k being

odd. Finally, consider that $(5^k - 2^y)(5^k - 2^y) = 3^{2t}$, we get $2^y = \frac{3^{2t}-1}{2} \equiv 1 \pmod 3$, so y is even.
(c) We know from the above discussion that $5^k - 2^y = 1$ and $5^k + 2^y = 3^{2t}$ (where $z = 2k$, $x = 2t$, k is odd). Consider modulo 8, y can be at most 2 (and must be even by (b)), it is easy to check that there is only one possible solution $(x, y, z) = (2, 2, 2)$.
6. (a) Use the Schröder-Bernstein Theorem to prove that $\mathbb{N}^m$ is denumerable for any $m \in \mathbb{N}$.
(Hint: For the function from $\mathbb{N}^m$ to $\mathbb{N}$, use the Fundamental Theorem of Arithmetic to help you to construct an injective function.)
(b) For any $m \in \mathbb{N}$, a function $f : \mathbb{Z} \longrightarrow \mathbb{N}$ is said to be periodic with period m if $f(x + m) = f(x)$ for any $x \in \mathbb{Z}$. Denote by P_m the set of all periodic functions $f : \mathbb{Z} \longrightarrow \mathbb{N}$ with period m. Use (a) and the Schröder-Bernstein Theorem to prove that P_m is denumerable for any $m \in \mathbb{N}$.
(Hint: First show that $|P_m| = |\mathbb{N}^m|$. Construct an injective function $\Phi : P_m \longrightarrow \mathbb{N}^m$ and an injective function $\Psi : \mathbb{N}^m \longrightarrow P_m$.)
Solution. (a) Straightforward if you follow the hint.
(b) Take $\Psi : \mathbb{N}^m \to P_m$, $(a_1, a_2, \cdots, a_m) \mapsto f_{a_1 a_2 \cdots a_m} : \mathbb{Z} \to \mathbb{N}$, $f_{a_1 a_2 \cdots a_m}(i) = a_i$ for $1 \le i \le m$. Then take $\Psi : P_m \to \mathbb{N}^m$, $f \mapsto (f(1), f(2), \cdots, f(m))$.